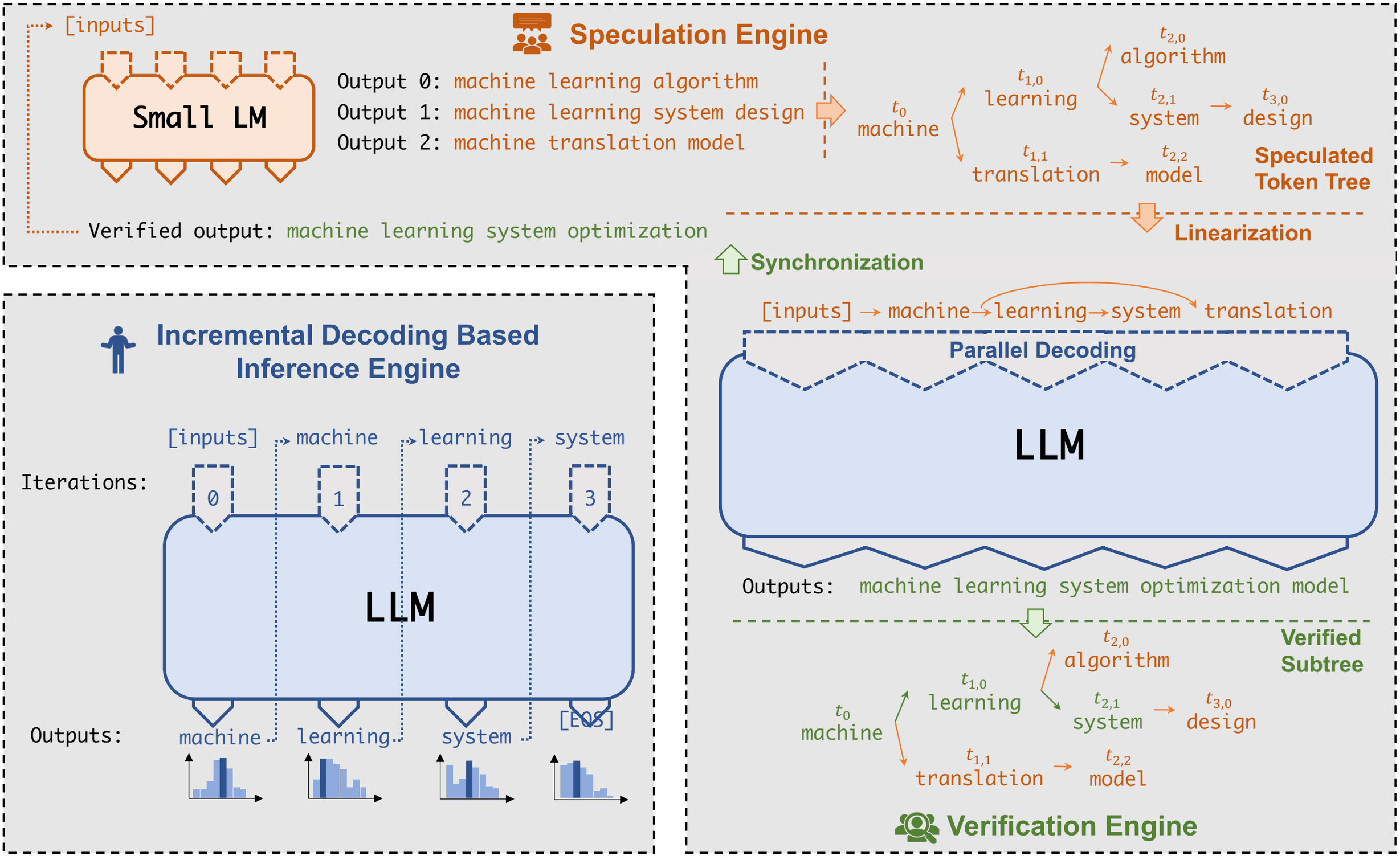
[inputs]
Small LM
Speculation Engine
Output 0: machine learning algorithm
Output 1: machine learning system design
Output 2: machine translation model
t_0 machine
$t_{1,0}$ learning
$t_{2,0}$ algorithm
$t_{2,1}$ system
$t_{3,0}$ design
$t_{1,1}$ translation
$t_{2,2}$ model
Speculated Token Tree
Verified output: machine learning system optimization
Linearization
Synchronization
[inputs] → machine → learning → system → translation
Parallel Decoding
LLM
Outputs: machine learning system optimization model
Verified Subtree
t_0 machine
$t_{1,0}$ learning
$t_{2,0}$ algorithm
$t_{2,1}$ system
$t_{3,0}$ design
$t_{1,1}$ translation
$t_{2,2}$ model
Verification Engine
Incremental Decoding Based Inference Engine
[inputs] machine learning system
Iterations:
0
1
2
3
LLM
Outputs:
machine
learning
system
[EOS]